E-commerce Product Delivery Prediction

UsingMachineLearningtoOptimizeLogisticsandImprove Customer Satisfaction

Problem & Objective

The Challenge

Late deliveries significantly reduce customer satisfaction and increase operational costs

Our Goal

Predict whether deliveries will reach customers on time using historical data

The Impact

Improve logistics planning, reduce delays, and enhance overall customer experience

Dataset Overview

Source File

E_Commerce.csv containing

comprehensiveshipping data

Target Variable

ReachedonTime(Binary: 1 = Yes, 0 = No)

Key Features

- Shipment Mode (Air, Ship, Truck)
- Customer Calls and Contacts
- Discount Applied
- Product Importance Level

Data Processing & EDA

Data Cleaning

Removedunnecessary columns and handled missing values

Feature Engineering

Convertedcategoricaldatato numeric format for model compatibility

Pattern Discovery

Usedgraphsandstatistical analysis to identify key delivery patterns

Models Used

1

Logistic Regression
Baseline model for binary classification

2

Decision Tree
Interpretable tree-based approach

3

Random Forest
Ensemble method with highest accuracy

4

K-Nearest Neighbors
Distance-based classification technique

Best Performing Model: Random Forest
Outperformed other algorithms in predicting on-time delivery status

Conclusion

Machine Learning Success

Successfullypredicted delivery status with high accuracy using Random Forest model

Operational Efficiency

Helpslogisticsteams improve delivery planning and resource alloc ation

Customer Satisfaction

Reduceslatedeliveriesandimproves overall customer experience

Questions ?

Thank you